SwiftHaul Logistics: Carrier and Lane Performance

Analytical summary, BigQuery analytics warehouse (Task 7)

Data: 1,500,000 shipments in `swifthaul.shipments` across 6 carriers, 56 lanes and 25 months, enriched with the carrier rate card (`carrier_rates_ext`). All queries are in `07_analysis.sql`; each figure below names the query and view it comes from.

Metrics: on-time rate = on-time shipments / total shipments, using the provided `on_time` flag. Cost per lane = average `freight_cost_usd` per origin and destination region pair. Cost per kg = average of `freight_cost_usd / weight_kg`.

1. Which carriers are the most and least reliable?

Source: Q1 in `07_analysis.sql`, from view `v_carrier_ontime_monthly`.

Carrier	Shipments	On-time rate	Worst month	Best month
Skyliner Air	250,025	**95.19%**	94.82%	95.58%
SwiftHaul Ground	250,269	93.17%	92.69%	93.59%
MetroCourier	249,964	92.14%	91.74%	92.50%
Continental Cargo	249,989	89.13%	88.69%	89.65%
BlueLine Freight	249,669	87.24%	86.21%	87.95%
OceanLink	250,084	**81.18%**	78.59%	81.86%

- **Most reliable: Skyliner Air (95.19%). Least reliable: OceanLink (81.18%)**, where almost 1 in 5 shipments is late.
- The ranking is stable. The monthly ranges do not overlap: every carrier's worst month is better than the next carrier's best month. For example, OceanLink's best month (81.86%) is below BlueLine Freight's worst month (86.21%).

2. Which lanes are the most expensive on average?

Source: Q2 (most expensive) and Q2b (cheapest) in `07_analysis.sql`, from view `v_lane_cost`.

Most expensive lanes	Avg cost	Cheapest lanes	Avg cost
APAC-East -> US-Central	**$403.31**	US-West -> US-Central	**$304.43**
US-West -> APAC-South	$402.96	US-West -> US-East	$304.83
EU-West -> APAC-South	$402.85	US-South -> US-Central	$305.44
APAC-East -> EU-Central	$402.41	APAC-South -> APAC-East	$305.78
US-East -> APAC-East	$402.20	APAC-East -> APAC-South	$306.40

- **The most expensive lane is APAC-East -> US-Central at $403.31 per shipment.** The top five lie within $1.11 of each other, so they are effectively tied.
- All five most expensive lanes are **intercontinental**, and all five cheapest are **within one continent**. The most expensive lane costs 32% more than the cheapest ($403.31 vs $304.43). Crossing continents drives lane cost, not any single lane. Each lane has roughly 26,500 to 27,000 shipments, so these averages are well supported.

3. How do carriers compare on cost vs reliability?

Source: Q3 in `07_analysis.sql`, from view `v_shipments_enriched` (shipments joined to `carrier_rates_ext` for mode).

Carrier	Mode	On-time rate	Avg cost per shipment	Avg cost per kg
OceanLink	Ocean	81.18%	$166.00	**$2.87**
SwiftHaul Ground	Ground	**93.17%**	$221.61	**$3.37**
BlueLine Freight	Ground	87.24%	$243.28	$3.51
Continental Cargo	Ground	89.13%	$279.28	$3.83
MetroCourier	Air	92.14%	$617.58	$6.48
Skyliner Air	Air	**95.19%**	$713.32	$7.31

- **SwiftHaul Ground is the best value.** It is the second most reliable carrier (93.17%) at the second lowest cost ($3.37 per kg).
- **Three carriers are outperformed on both measures by SwiftHaul Ground.** BlueLine Freight, Continental Cargo and MetroCourier are each more expensive per kg *and* less reliable. MetroCourier costs almost twice as much per kg ($6.48 vs $3.37) and is still 1.0 point less punctual.
- **The real trade-off is between three carriers:**
    - **Skyliner Air** buys 2.0 extra points of on-time rate (95.19% vs 93.17%) for 2.2 times the cost per kg ($7.31 vs $3.37).
    - **OceanLink** saves 15% per kg against SwiftHaul Ground ($2.87 vs $3.37), but its on-time rate is 12.0 points lower (81.18% vs 93.17%).

Recommendation

1. **Make SwiftHaul Ground the default carrier.** It gives near-top reliability at close to the lowest cost.
2. **Reserve Skyliner Air for shipments where a late delivery is costly**, such as Express orders or high-value customers. Its 2-point reliability gain does not justify 2.2 times the price for routine freight.
3. **Use OceanLink only for freight that is not time-sensitive**, and only where the 15% saving outweighs a late rate of about 19%.
4. **Reduce volume with BlueLine Freight, Continental Cargo and MetroCourier.** SwiftHaul Ground beats each of them on both cost and reliability. Use their contract renewals to renegotiate rates or move the volume.

Limitations: `on_time` measures delivery against each shipment's service-level window, and these queries do not control for the mix of service levels (Economy, Standard, Express) or mode capabilities per carrier. Before moving volume, a carrier by service level breakdown should confirm that the ranking holds within each service level.